1.3 Continuity

AP Calculus 1.3 · Student sheet

Name _____ Period ____ Date _____

I can…

- decide whether a function is continuous at a number using the three conditions.
- find the intervals where a function is continuous and remove a removable discontinuity.
- use the Intermediate Value Theorem to show that a function has a zero.

Vocabulary · fill in the missing word

Continuous at a number c — f(c) is _____, the limit of f(x) as x → c exists, and the limit equals f(c)

Removable discontinuity — a discontinuity at c where the limit exists, so it can be removed by defining or _____ f(c)

Intermediate Value Theorem — if f is _____ on [a, b] and N is between f(a) and f(b), then f(c) = N for at least one c in (a, b)

Part 1 · The pencil test

For each graph near x = c, check each condition.

Graph	f(c) defined?	Limit exists?	Limit = f(c)?	Pencil test?
Smooth				
Hole				
Moved point				
Jump				

Part 2 · Notes and examples (fill in each note, then work the example)

Continuity at c: f(c) is defined, the limit as x → c exists, and the limit equals _____.

Example 1

f(x) = 3x − 1 if x < 1; 4 if x = 1; 4x if x > 1. Is f continuous at 1?

Removable discontinuity: the limit exists, so the discontinuity can be removed by defining or _____ f(c).

Example 2

$f(x) = \dfrac{x^2 - 9}{x - 3}$. Where is f discontinuous, and how can the discontinuity be removed?

Example 3

f(x) = 2x + 1 if x ≤ 2; $kx^2 - 3$ if x > 2. Find k so that f is continuous at 2.

Continuous functions: polynomials are continuous everywhere; rational functions are continuous on their _____.

Example 4

Where is $R(x) = \dfrac{x^2 - 4}{x^2 - 9}$ continuous? Which discontinuities are removable?

Intermediate Value Theorem: if f is continuous on [a, b] and N is between f(a) and f(b), then f(c) = _____ for at least one c in (a, b).

Example 5

Show that $f(x) = x^3 + 3x - 5$ has a zero in [1, 2].

Part 3 · Partner practice

Partner 1

$f(x) = x^2$ if $x < 1$; 2 if $x = 1$; $2x - 1$ if $x > 1$. Is f continuous at 1? If not, what kind of discontinuity is it?

Partner 2

(AP® Practice) For what value of k is $f(x) = kx + 1$ if $x \le 3$; $x^2 - 2$ if $x > 3$ continuous at $x = 3$?

(A) 1 (B) 2 (C) $\dfrac{7}{3}$ (D) 3